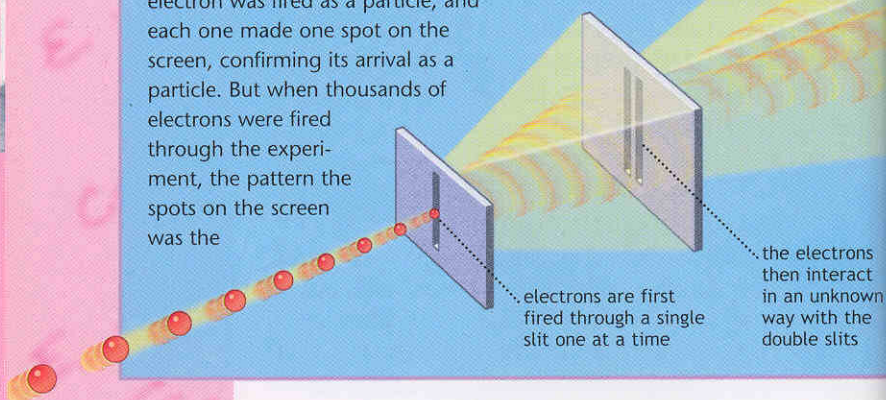


the heart of the matter

Quantum entities behave as if they are both wave and particle. In an experiment in the 1980s, electrons were fired, singly, through the equivalent of Young's double-slit experiment. They were detected by a screen like a TV screen, where each one made a spot of light that the screen "remembered." Each electron was fired as a particle, and each one made one spot on the screen, confirming its arrival as a particle. But when thousands of electrons were fired through the experiment, the pattern the spots on the screen was the

classic interference pattern for waves. Exactly equivalent experiments have also been done with photons (particles of light). You don't have to understand this—nobody understands it. You just have to accept that this is the way the quantum world works.

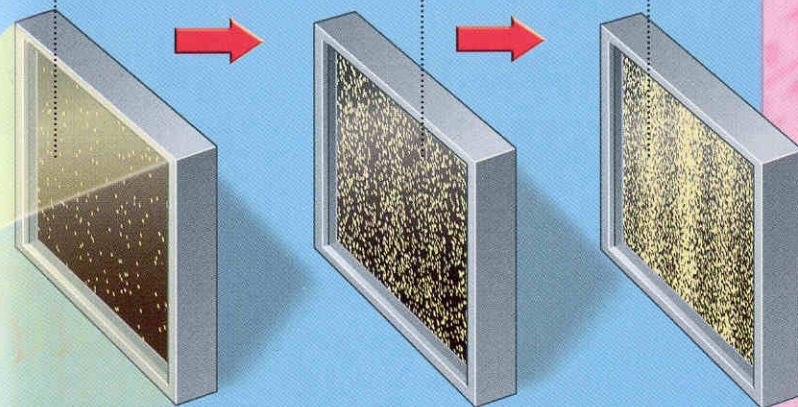


Each of the electrons made a single spot of light, proving that they are particles. As the electrons were fired one after another, the spots of light gradually built up a clear pattern—the diffraction pattern—not the two "heaps of rocks." The electrons were not even traveling through the experiment together, but they still built up the characteristic pattern associated with waves passing through the two slits. It was as if each electron knew where all its predecessors had gone, and where all the electrons coming behind would go, as well as knowing its own place in the pattern. Nothing demonstrates the weirdness of the quantum world more clearly to nonscientists.

single electrons register on the screen as spots of light

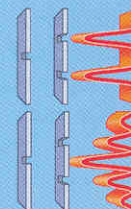
the electrons gradually accumulate with an emerging pattern.

finally, the interference pattern that is characteristic of waves appears.



the particle wave

The experiment only allows one electron at a time to pass through the apparatus. But the pattern that builds up can only be produced by waves that have passed through both slits and interfered with each other. It isn't even the pattern you would expect if separate waves went through one or the other of the slits one at a time.



what you think it should look like

how the actual pattern appears

Of course, the electron does not "know" anything. It is simply following the blind rules of chance in the same way that a true die does not "know" what numbers have come up previously (or are coming next) in order for the probability of rolling, for example, a three next time to be exactly one in six. Probability runs very deep in quantum physics, something Einstein hated, famously commenting, "I cannot believe that God plays

roll of the dice
A tumbling die lands with one facing up in accordance with the rules of probability. It doesn't have to "know" what number came up last time. Quantum processes also obey probabilistic rules.

